General Calculus II

Course Text

- This course does not require a text.

Course Description and Objectives

This course is designed to further acquaint students with the principles of Calculus. This includes techniques of integration; application of integration; exponential and logistic models; parametric equations and polar coordinates; sequence and series; and vector and geometry. Upon successful completion of the course, students will be able to:

- Understand other Intermediate forms and how to solve the problems with different intermediate forms.
- Understand hyperbolic function and hyperbolic identities, learn how to find derivative of hyperbolic function.
- Understand techniques of Integration and learn how to solve - Integration using table and u-substitution, integration by partial fraction, integration by using trigonometric substitution, how to solve numerical integration.
- Learn applications of integral – understand the average value of function, understand how to find - volume of revolution, surface of revolution and arc length of functions.
- Understand Sequences and Series -learn monotonic, bounded sequences and indefinite series. Understand how to check convergence and divergence of series, solve problems based on Taylor and McLaurin series and convergence and divergence of power series.
- Understand what differential equation is, learn how to solve Homogeneous differential equations, and solve growth and decay problems.
- Understand what are parametric equations and polar coordinates.
- Understand vectors.

The topics covered under this course are other indeterminate forms, the hyperbolic functions; the techniques of integration; application of integral calculus; sequences and series; differential equations; parametric equations and polar coordinates; and vectors and geometry.

Course Prerequisites

StraighterLine does not require prerequisites, however it is highly recommended that students take General Calculus I or its equivalent before enrolling in General Calculus II. Concepts learned in General Calculus I are necessary in order to successfully complete General Calculus II.

Important Terms

In this course, different terms are used to designate tasks:

- **Proctoring**: all final exams require proctoring which can be completed conveniently from your home. A webcam is required.
- **Tutoring**: memberships include online tutoring for students to access with any content/subject related questions in the place of faculty. If your tutor is not able to

answer your questions please contact a student advisor.
- **Exam**: A graded online test.
- **Exercises**: ungraded practice exercises and quiz questions.

Course Evaluation Criteria

StraighterLine provides a percentage score and letter grade for each course. See Academic Questions section in FAQ for further details on percentage scores and grading scale. A passing percentage is **70%** or higher.

If you have chosen a Partner College to award credit for this course, your final grade will be based upon that college's grading scale. Only passing scores will be considered by Partner Colleges for an award of credit.

There are a **total of 1000 points** in the course:

Chapter	Assessment	Points Available
4	Graded Exam 1	125
6	Graded Exam 2	125
7	Midterm Exam	200
9	Graded Exam 3	125
11	Graded Exam 4	125
12	Final Exam	300
Total		1000

Course Topics and Objectives

Chapter	Topics	Subtopics
An Introduction to Calculus II	• Introduction	• Welcome to Calculus II • Review: Calculus I in 20 minutes
Math Fun	• Paradoxes • Sequences	• An Introduction to Paradoxes • Paradoxes and Air Safety • Newcomb's Paradox • Zeno's Paradox • Fibonacci Numbers • The Golden Ratio

Other Indeterminate Forms	• Indeterminate Form $0 \cdot \infty$, $\infty - \infty$ 1^{∞}, 0^{0}, ∞^{0}	• L'Hopital's rule and Indeterminate Products • L'Hôpital's rule and Indeterminate Differences • L'Hôpital's rule and One to the Infinite Power • Another example of One to the Infinite Power • L'Hôpital's rule and zero to the zero power • L'Hôpital's rule and infinity to the zero power
The Hyperbolic Functions	• Hyperbolic Functions	• Defining the Hyperbolic Functions • Hyperbolic Identities • Derivatives of Hyperbolic Functions
Techniques of Integration	• Integration Using Tables • Integrals Involving Powers of Sine and Cosine • Integrals Involving Powers of Other Trigonometric Functions • Integration by Partial Fractions and Repeated Factors • An Introduction to Trigonometric Substitution • Trigonometric Substitution Strategy • Numerical Integration	• An Introduction to the Integral Table • Making u-Substitutions • An Introduction to Integrals with Powers of Sine and Cosine • Integrals with Powers of Sine and Cosine • Integrals with Even and Odd Powers of Sine and Cosine • Integrals of Other Trigonometric Functions • Integrals of Odd Powers of Tangent and Any Power of Secant • Integrals with Even Powers of Secant and Any Power of Tangent • Repeated Linear Factors: Part One • Repeated Linear Factors: Part Two • Distinct and Repeated Quadratic Factors • Partial Fractions of Transcendental Functions • Converting Radicals into Trigonometric Expressions • Using Trigonometric Substitution to Integrate Radicals • Trigonometric Substitutions on Rational Powers • An Overview of Trigonometric Substitution Strategy • Trigonometric Substitution Involving a Definite Integral: Part One • Trigonometric Substitution Involving a Definite Integral: Part Two • Deriving the Trapezoidal Rule • An Example of the Trapezoidal Rule

Applications of Integral Calculus	• The Average Value of a Function • Finding Volumes Using Cross-Sections • Disks and Washers • Shells • Arc Lengths and Functions • Surface of Revolution • Work • Moments and Centers of Mass	• Finding the Average Value of a Function • Finding the Volumes Using Cross-Sectional Slices • An Example of Finding Cross-Sectional Volumes • Solids of Revolution • The Disk Method along the y-Axis • A Transcendental Example of the Disk Method • The Washer Method across the x-Axis • The Washer Method across the y-Axis • Introducing the Shell Method • Why Shells Can Be Better Than Washers • The Shell Method: Integrating with Respect to y • An Introduction to Arc Length • Finding Arc Lengths of Curves Given by Functions • Finding Area of a Surface of Revolution • An Introduction to Work • Calculating Work • Hooke's Law • Center of Mass • The Center of Mass of a Thin Plate
Sequences and Series	• Sequences • Monotonic and Bounded Sequences • Infinite Series • Convergence and Divergence • The Integral Test and p-Series • The Direct Comparison Test • The Limit Comparison Test	• The Limit of a Sequence • Determining the Limit of a Sequence • Monotonic and Bounded Sequences • An Introduction to Infinite Series • The Summation of Infinite Series • Geometric Series • Telescoping Series • Properties of Convergent Series • The nth-Term Test for Divergence • An Introduction to the Integral Test • Examples of the Integral Test • Using the Integral Test • Defining p-Series • An Introduction to the Direct Comparison Test • Using the Direct Comparison Test • An Introduction to the Limit Comparison Test • Using the Limit Comparison Test • Inverting the Series in the Limit Comparison Test
Sequences and Series (continued)	• The Alternating Series	• Alternating Series • The Alternating Series Test

	• Absolute and Conditional Convergences • The Ratio and Root Test • Polynomial Approximations of Elementary Functions • Taylor and Maclaurin Polynomials • Taylor and Maclaurin Series • Power Series • Power Series Representations of Functions	• Estimating the Sum of an Alternating Series • Absolute and Conditional Convergence • The Ratio Test • Examples of the Ratio Test • The Root Test • Polynomial Approximations of Elementary Functions • Higher-Degree Approximations • Taylor Polynomials • Maclaurin Polynomials • The Remainder of a Taylor Polynomial • Approximating the Value of a Function • Taylor Series • Examples of the Taylor and Maclaurin Series • New Taylor Series • The Convergence of Taylor Series • The Definition of Power Series • The Interval and Radius of Convergence • Finding the Interval and Radius of Convergence: Part One • Finding the Interval and Radius of Convergence: Part Two • Finding the Interval and Radius of Convergence: Part Three • Differentiation and Integration of Power Series • Finding Power Series Representations by Differentiation • Finding Power Series Representations by Integration • Integrating Functions Using Power Series
Differential Equations	• Solving a Homogeneous Differential Equation • Growth and Decay Problems	• Separating Homogeneous Differential Equations • Example of Newton's Law of Cooling • Change of Variables • Exponential Growth • Logistic Growth • Radioactive Decay
Parametric Equations and Polar Coordinates	• Understanding Parametric Equations • Calculus and Parametric Equations • Understanding Polar Coordinates • Polar Functions and Slope	• An Introduction to Parametric Equations • Sketching a Parametric Curve • The Cycloid • Eliminating Parameters • Derivatives of Parametric Equations • Finding the Slopes of Tangent Lines in Parametric Form

	• Polar Functions and Area	• Graphing the Elliptic Curve • The Arc Length of a Parameterized Curve • Finding Arc Lengths of Curves Given by Parametric Equations • The Polar Coordinate System • Converting between Polar and Cartesian Forms • Spirals and Circles • Graphing Some Special Polar Functions • Calculus and the Rose Curve • Finding the Slopes of Tangent Lines in Polar Form • Heading toward the Area of a Polar Region • Finding the Area of a Polar Region: Part One • Finding the Area of a Polar Region: Part Two • The Area of a Region bounded by Two Polar Curves: Part One • The Area of a Region bounded by Two Polar Curves: Part Two • The Arc Length of a Polar Curve • Area of surface of revolution in Polar Form
Vectors and the Geometry of R^2 and R^3	• Vectors and the Geometry of R^2 and R^3 • Vector Functions	• Coordinate Geometry in Three Dimensional Space • Introduction to Vectors • Vectors in R^2 and R^3 • An Introduction to the Dot Product • Orthogonal Projections • An Introduction to the Cross Product • Geometry of the Cross Product • Equations of Lines and Planes in R^3 • Introduction to Vector Functions • Derivatives of Vector Functions • Vector Functions: Smooth Curves • Vector Functions: Velocity and Acceleration
Review and Final Exam	Review and Final Exam	• Review and Final Exam